

Theory Lectures on Manufacturing Process (NMES 101)

Solidification time



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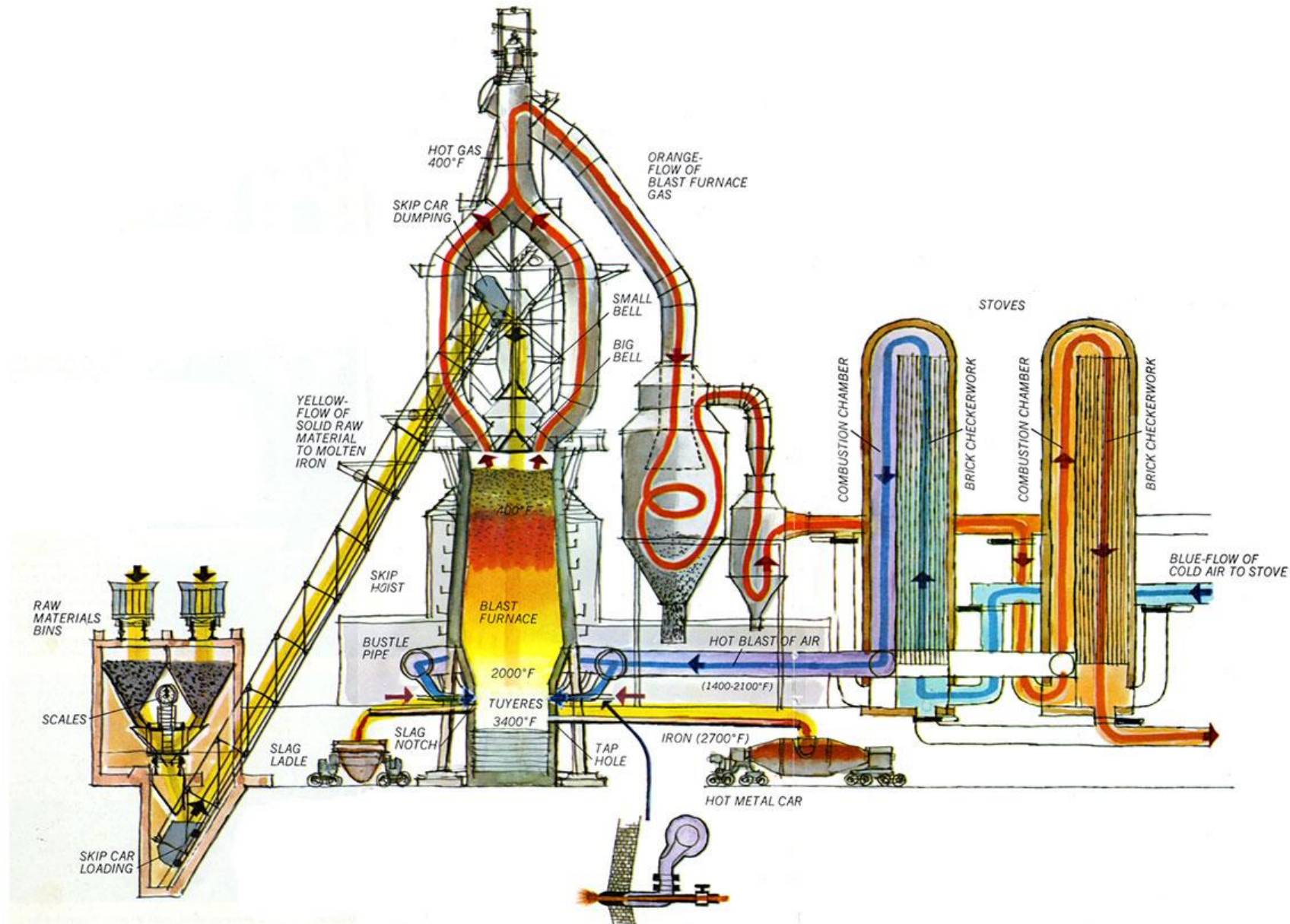
Content prepared for theory lecture for
1st Year BTech level students for common
course on Manufacturing Processes.
Each lecture duration $\approx$ 45 min



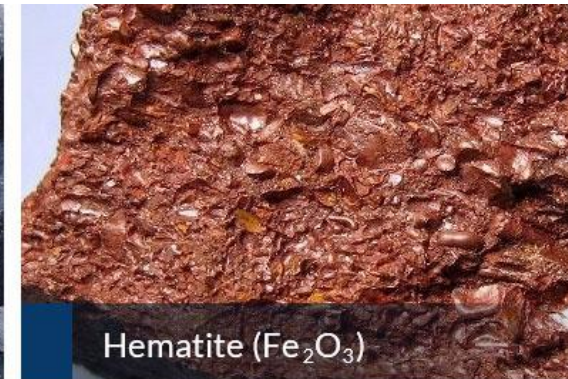
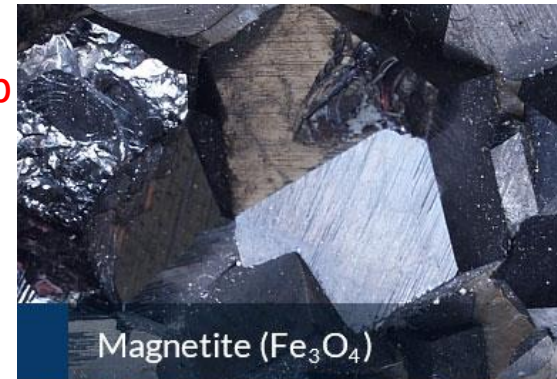
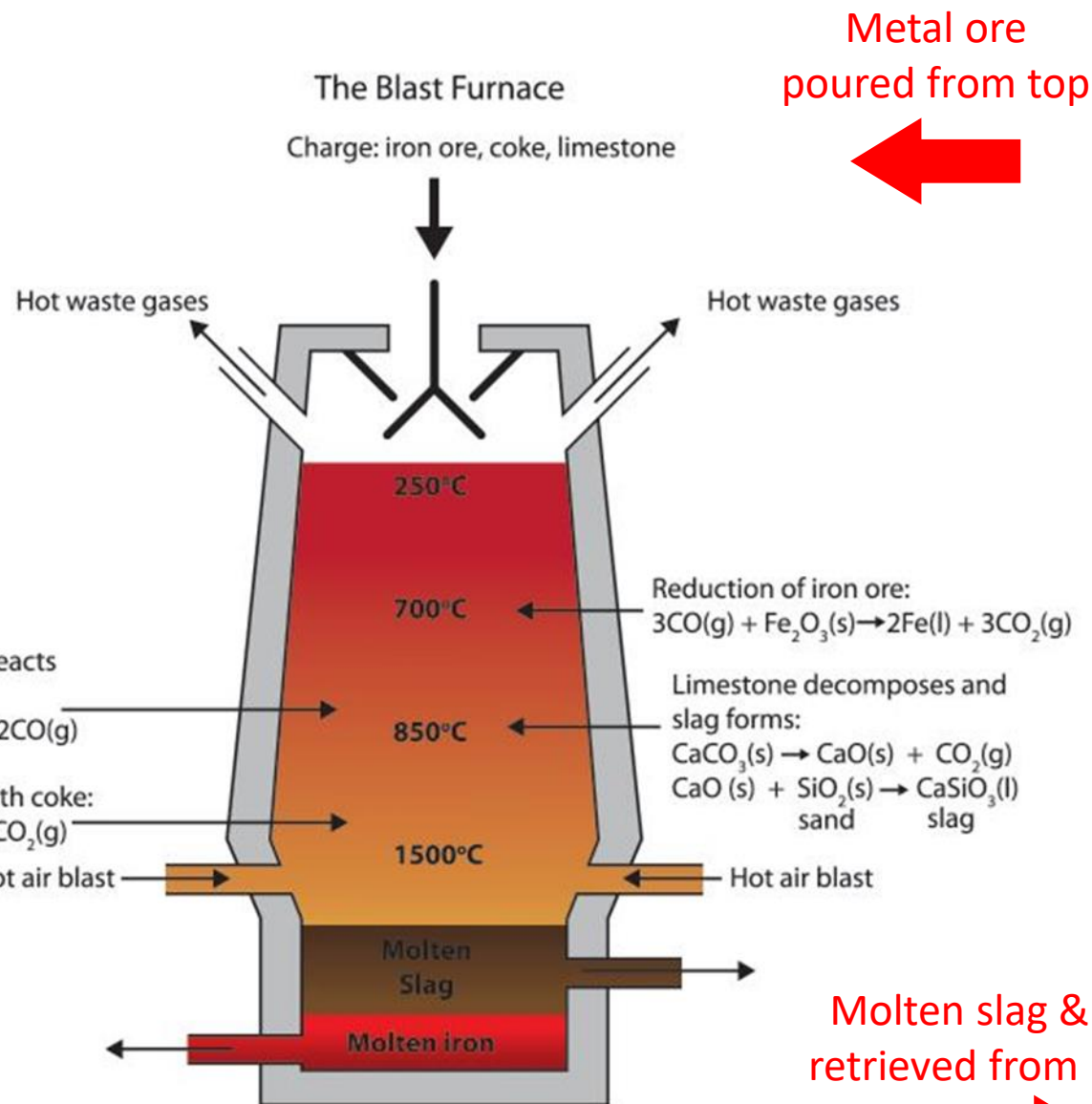
Steel Melting – Blast Furnace

IISCO Steel Plant (ISP), Burnpur
Steel Authority of India Limited (SAIL)





Steel Melting Shop (SMS) – blast furnace



Molten slag & metal retrieved from bottom



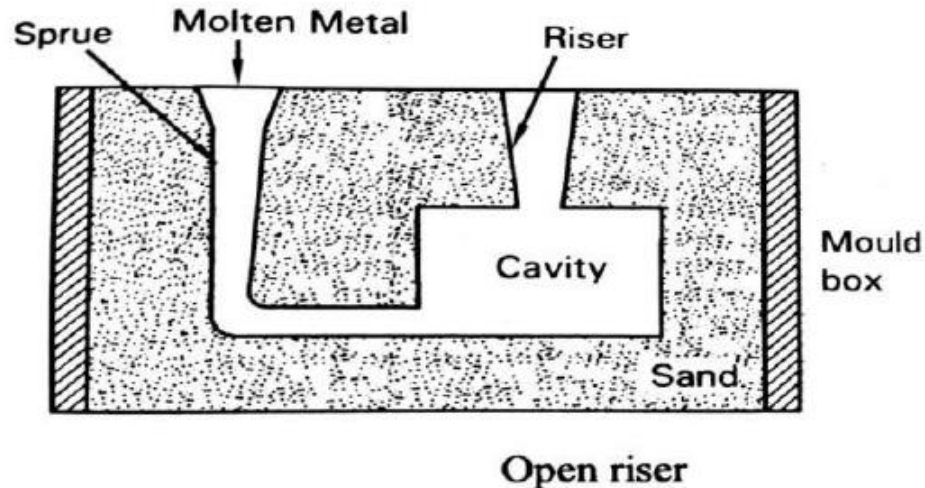
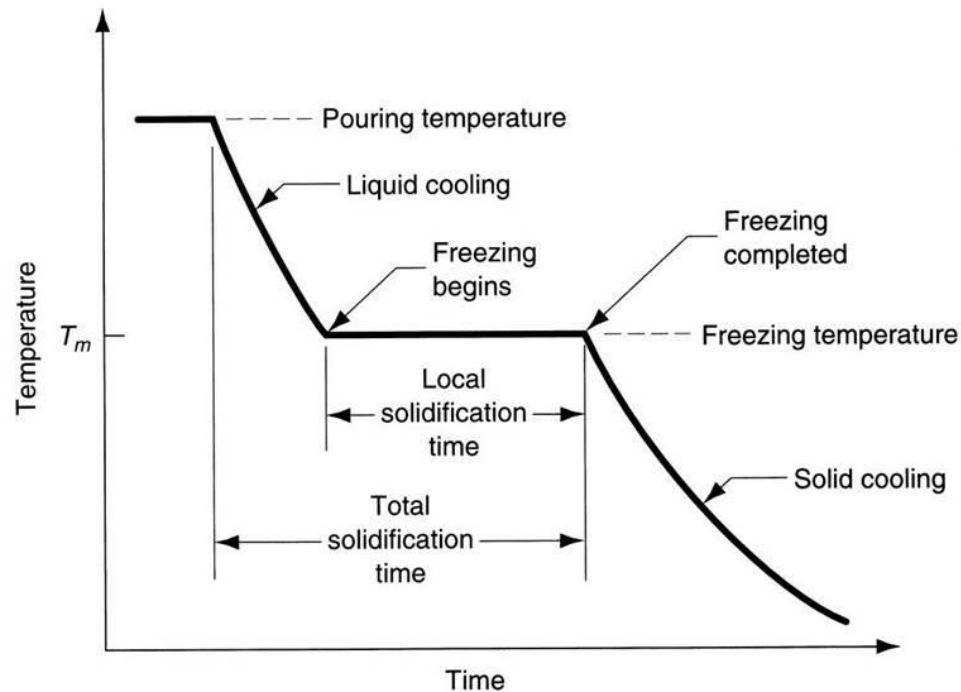
Ladle and Pouring of molten metal into cast cavity



© Bob Richardson



Shape and inner volume (m³), calculated from taphole level above and 1 m below charging device								Working height Inner height Total height (H) Tuyere Taphole
Hearth diameter (mm)	900	4500	6500	9000	9750	11,200	14,000	
Total height H (mm)	15,300	20,300	24,000	26,075	29,400	31,450	36,000	
Productivity: THM/day	25	450	1200	>2500	4000	6000	~10,000	



During the solidification process, volumetric shrinkage takes place. There are three stages of shrinkage:

1. **Liquid stage** shrinkage (Riser takes care of it)
2. Shrinkage during cooling of **solid metal** (Shrinkage allowance takes care of it)

➤ Functions of the Riser

1. Liquid metals and their alloys shrink as they cool and solidify. It creates a partial vacuum within casting resulting in the formation of shrinkage void or cavity if extra metal from outside the mold (cavity) is not supplied. Riser stores the molten metal within itself during pouring of molten metal into cavity. Then this stored metal flows from riser to the cavity to compensate for the volumetric loss during **liquid phase shrinkage**. So, the primary function of the riser (attached with mold) is to supply molten metal to the solidifying casting (cavity) so that shrinkage cavities are get formed. (**Shrinkage during solid stage** is taken care of by **Shrinkage Allowance**).
2. Riser also **permits the escape of air and mold gases** when the mold cavity is being filled with the molten metal.
3. A riser full of molten metal **indicates that the mold cavity has already been completely filled up** with the same.



Chvorinov's Rule of solidification time



Same surface area but different volume
Which one will cool down quickly?

The higher is the volume of the cast product, the longer will be the solidification time.

- Solidification time (t) is **proportional** to the square of the cast volume (V).

$$t \propto V^2$$



Which one will cool down quickly?

The higher is the surface area of the cast product, the shorter will be the solidification time.

- Solidification time is **inversely proportional** to the square of the cast surface area (A).

$$t \propto \frac{1}{A^2}$$

Chvorinov's Rule of solidification time

$$\therefore t \propto \left(\frac{V}{A} \right)^2$$



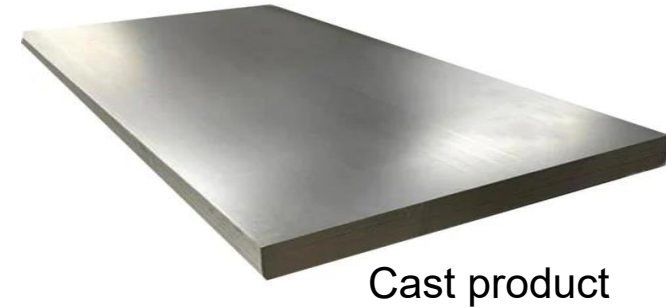
Problem: An aluminum plate of size 125 mm x 125 mm x 25 mm is to be sand cast. Find out the solidification time using Chvorinov's Rule with constant of proportionality as 2.

- Volume of casting, $V_c = 125 \times 125 \times 25 = 390625 \text{ mm}^3$
- Surface area of casting, $A_c = (2 \times 125 \times 125) + (4 \times 125 \times 25) = 43750 \text{ mm}^2$
- Proportionality constant, $k = 2$.
- Apply Chvorinov's Rule: $t = 2 \times (390625 / 43750)^2$
- $t = 159 \text{ min}$



Problem: An open cylindrical riser must be designed for a sand mold. The part to be cast is a 125 mm x 125 mm x 25 mm plate. It is required that the height-to-diameter ratio of the riser be 1. Find the dimensions of the riser so that its total solidification time is 30% longer than the casting.

- Volume of casting, $V_c = 125 \times 125 \times 25 = 390625 \text{ mm}^3$
- Surface area of casting, $A_c = (2 \times 125 \times 125) + (4 \times 125 \times 25) = 43750 \text{ mm}^2$
- For defect-free casting it is required that $t_{\text{riser}} \geq t_{\text{casting}}$
- Problem states that $t_{\text{riser}} = 1.3 t_{\text{casting}}$
- Riser shape is cylindrical
- Height-to-diameter ratio of the riser must be 1. So $\Rightarrow D = h$
- Volume of riser, $V_r = \pi D^2 h / 4 = \pi D^3 / 4$
- Surface area of riser, $A_r = (\pi D h) + 1 \times (\pi D^2 / 4)$ [Heat transfer is not possible from the circular end that is in contact with case product]
- Chvorinov's Rule: $(V_r / A_r)^2 = 1.3 (V_c / A_c)^2$
- **$D = h = 50.74 \text{ mm}$**





Problem 1: A cylindrical riser must be designed for a sand-casting mold. The casting itself is a steel rectangular plate with dimensions 7.5 cm x 12.5 cm x 2.0 cm. Previous observations have indicated that the total solidification time for this casting = 1.6 min. The cylinder for the riser will have a diameter-to-height ratio = 1.0. Determine the dimensions of the riser so that its total solidification time = 2.0 min

Problem 2: A cylindrical riser needs to be designed to produce a cylindrical product by sand casting. The radius of the riser is same with its height. The cylindrical cast cavity has 300 mm diameter and 60 mm length with axis in vertical direction. Riser is coaxially placed at the top of the cast cavity. Determine the radius of the cylindrical riser so that the solidification time of riser is 1.2 times that of cast metal.

Problem-3: A cubic riser needs to be designed for a sand mould. The part to be cast is a 150 mm × 150 mm × 25 mm plate. Find the dimensions of the riser so that its total solidification time is 30% longer than the casting.

Problem-4: A cube-shaped riser needs to be designed to produce a cylindrical product by sand moulding. Thus, a cylindrical cast cavity of 100 mm diameter and 60 mm length is prepared with axis in vertical direction. Riser is placed at the top of the cavity. Determine the dimension of the cube-shaped riser so that the solidification time of riser is 1.2 times higher than that of cast metal.